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A cost effective degradation-based maintenance strategy under imperfect repair

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ABSTRACT

An optimization model is developed to minimize the total cost of imperfect degradation-based maintenance by determining an optimal interval of condition monitoring and the degradation level after imperfect preventive repairs. The decision model is based on a novel cost model that considers functional relationship between the expected degradation reduction and the cost of preventive repairs. The decision model is applied to simulated vibration signals with a variety of specifications of cost values and degradation model parameters. This study has initiated a new area for the research of cost effective maintenance strategies. The results clearly indicate the significance of the proposed model and the decision variables under the objective of minimal cost. For instance, the results indicate direct relationship between the optimal length of monitoring interval and the monitoring cost. However, longer monitoring interval increases the risk of failure, and therefore, more degradation reduction is needed. By increasing the slope of cumulative degradation, the cost effective strategy advocates taking more frequent monitoring. The optimal degradation level after each preventive repair is not so sensitive to the change in the degradation slope due to the uncertainty associated with degradation patterns.

1. Introduction

In recent years, condition-based maintenance (CBM) has been widely accepted as the most efficient maintenance strategy [1–4]. Under the CBM philosophy, a system is maintained only when there is an objective evidence of impending failures. Degradation-based maintenance (DBM) is a form of CBM which relies on monitoring and analysis of degradation data [5–7]. Variables such as fatigue cracks, vibration features, temperature, and oil concentration are used as degradation indicators and reflect the actual deterioration status in real-time [8–10]. In essence, the degradation mechanisms can be related to life-time distributions through degradation rate or cumulative degradation models. A variety of degradation models exist in the literature including Wiener process models [11], Gamma process [12], inverse Gaussian process [13], general path models [14] and random coefficient models [15]. A comprehensive review of degradation models is given by Ye and Xie [16].

The concept of imperfect maintenance, introduced by Brown and Proschan [17], implies that the system condition after maintenance actions is somewhere between good-as-new and bad-as-old. There has been growing interest in studying health management under the realistic assumption of imperfect maintenance. For instance, Zhao et al. [18] proposed a model to optimize the intervals of imperfect and non-periodic inspections for a multi-defect component. Sanches et al. [19] considered uncertainties in the modelling of imperfect maintenance to optimize availability and maintenance cost. An optimal replacement strategy was proposed by Liu and Huang [20] for a multi-state system incorporating imperfect maintenance to determine the optimal failure number before replacing the entire system. Pandey et al. [21] addressed the optimal maintenance decision for binary systems to maximize the next mission reliability under imperfect maintenance. For extensive discussions regarding imperfect maintenance models readers may refer to, Shafiee and Chukova [22], and Zhang et al. [7].

More attention has been paid to imperfect DBM in recent years. Zhou et al. [23] proposed a maintenance scheduling model based on optimized cumulative maintenance cost for a system subject to degradation. Under budget constraints, Nourelfath et al. [24] aimed to maximize the system availability by jointly determining redundancy and imperfect preventive maintenance (PM) in series–parallel multi-state degraded systems. Chen [25] developed an optimal schedule for imperfect PM to minimize the expected cost rate for a system experiencing both minor and catastrophic shocks. Wang and Pham [26] considered the dependency between failures due to random shocks and degradation processes for one single-unit system. Zhang et al. [7] proposed a model based on the fact

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Notations

X_p	degradation threshold for potential failure
X_f	degradation threshold for functional failure
ζ	optimal degradation monitoring interval
x_r	degradation level after repair
$D(t_i)$	cumulative degradation model up to time t_i
Φ	initial degradation value (a known constant)
$\varepsilon(t_i)$	error term that follows a Markov process
θ	lognormal random variable
β	normal random variable
$L(t_i)$	log form of the cumulative degradation to time t_i
Δ_i	increment of log degradation
C_P	cost of preventive repairs
Cs	fixed (set up) cost for preventive repairs
M	proportional constant (known)
D_k	expected cumulative degradation just before the kth preventive repair
U_k	expected cumulative degradation immediately after the kth preventive repair
r_k	reduction factor of cumulative degradation for the kth preventive repair
$L(t_k)$	log cumulative degradation right before preventive repairs at time t_k
α	nonlinearity factor
$LC(\zeta, x_r)$	expected length of life cycle
$TC(\zeta, x_r)$	expected total cost of maintenance during a life cycle
C_f	mean cost per failure and replacement
C_m	mean cost per monitoring or inspection
N_p	number of preventive repairs within a life cycle
N	number of condition monitoring or inspections within a life cycle
k	number of preventive maintenance
T	set of monitoring times
$C(\zeta, x_r)$	expected life cycle maintenance cost per unit time
$P_f(x, \zeta)$	probability that a failure occurs before preventive repairs
$P_f(i, x, \zeta)$	probability that a preventive repair occurs before the failure

that imperfect maintenance may change the rate of degradation rather than the level of degradation. Liu et al. [27] performed an outstanding research to develop PM policy for multi-component-system with an objective to maximize the maintenance net value. To this end, it is to be noted that considerable research has been devoted to optimize maintenance costs. In this study, we present a novel way of dealing with cost effective maintenance strategies.

The objective of this research is to minimize the total cost of an imperfect DBM strategy by determining the length of the monitoring interval and the degradation level after imperfect preventive repairs. In order to achieve our objective, a model is provided for the cost of imperfect DBM under periodic monitoring using a non-constant cost of preventive repairs. In this respect, the functional relationships between degradation reduction and cost of preventive repairs for imperfect DBM is discussed. Furthermore, a decision model is developed based on the proposed cost model. The decision model is applied to vibration health monitoring with a variety of specifications of cost values and degradation model parameters.

The remainder of the paper is organized as follows. Section 2 is devoted to the degradation process modeling. Section 3 presents an imperfect DBM cost model. In this section, we introduce the functional relationship between the expected degradation reduction and the cost of preventive repairs. Next, the probabilities of interest are derived to determine the expected life cycle length and the expected total cost of maintenance during a life cycle. These expected values are combined to derive the maintenance cost model. The cost model is used in Section 4 to propose a decision model that minimizes the life cycle cost. Section 5 provides a numerical example. In Section 6, the proposed methodology is applied to a group of simulated vibration signals. Finally, concluding remarks are presented in Section 7.

2. Degradation process modeling

2.1. Defining thresholds

The threshold between normal state and potential failure is denoted by X_p. In other words, X_p is the threshold for initiating imperfect preventive repairs. The threshold between potential failure and functional failure is denoted by X_f. Hence, if an indicator for the measurable degradation process is between X_p and X_f, then a potential failure is indicated and necessary preventive repairs may be required. If the degradation indicator exceeds X_f, a functional failure is assumed to occur and perfect maintenance, i.e. replacement, needs to be carried out.

In essence, the cost effectiveness of maintenance actions depends on the values of the above-mentioned thresholds. In many studies, the degradation threshold for potential failure is assumed to be pre-determined [28,29]. In this research, the thresholds are log transformations of original degradation values. Assuming the availability of pre-determined critical thresholds, two decision variables are used to minimize the life cycle cost: (i) the optimal length of monitoring interval, ζ; (ii) the degradation level after each preventive repair, x_r. It is conceivable that after each imperfect repair such as balancing or alignment, the degradation level drops to a fixed value (x_r) between the initial condition and X_p. We assume that the degradations before and after repair are independent.

2.2. Degradation model

An appropriate degradation model representing the functional form of degradation is needed to predict the state of system under monitoring. This research models degradation using an exponential model proposed by Nelson [30] in which the error term follows a Markov process with independently distributed increments. Let $D(t)$ denote the cumulative degradation up to time t. At time t_i the cumulative degradation can be expressed as the following exponential form with mutually independent parameters θ and β

$$D(t_i) = \Phi + \theta\, e^{\beta t_i + \varepsilon(t_i)} \quad i = 1, 2, \ldots; \qquad 0 \le t_1 \le t_2 \le \ldots \tag{1}$$

where $\ln\theta$ has a normal distribution, and β is a normal random variable. The error term, $\varepsilon(t_i)$, follows a Markov process. By log transformation, the log form of the exponential model in Eq. (1) is given by

$$L(t_i) = \ln(D(t_i) - \Phi) = \ln\theta + \beta t_i + \varepsilon(t_i). \tag{2}$$

where θ and β represent characteristics common to all individual systems in the population. Stochastic parameters in error terms are used to represent the degradation characteristics unique to an individual system. By the Markovian property of error terms, any two increments in log degradation are independent from each other and so are the increments in original degradation values. Let $L(t_0) = 0$ and $\theta' = \ln\theta$. Hence, the degradation model in a random coefficient form is given by

$$L(t_i) = \theta' + \beta t_i + \varepsilon(t_i). \tag{3}$$

It is obvious that the log degradation accumulates continually over time and follows a random growth process where the mean values of the observed condition measures are functions of time.

3. Maintenance cost model

3.1. Model parameters

To define the expected cost of preventive repairs, it is necessary to consider the relationship between degradation reduction and cost of repair. In essence, this relationship might be different for specific applications. A linear relationship between degradation reduction and cost of a preventive repair can be given as

$$C_p = M(1-r_k)D_k + C_s \tag{4}$$

A non-linear relationship between degradation reduction and the cost of preventive repairs is

$$C_p = M(1-r_k)^{\alpha} D_k + C_s \tag{5}$$

For both models, the reduction factor is $0 < r_k < 1$. The only difference between two models is the nonlinearity factor, $\alpha > 1$. Considering the degradation model and the degradation level after each preventive repair, the linear relationship between degradation reduction by imperfect maintenance and the cost of a preventive repair, at monitoring i can be written as

$$C_p[i] = M(L(t_{i-1}) - x_r) + C_s \tag{6.a}$$

Similarly, the non-linear relationship between degradation reduction and cost of preventive repairs is

$$C_p[i] = M(L(t_{i-1}) - x_r)^{\alpha} + C_s \tag{6.b}$$

The above two equations provide the variable cost of PM required to determine the expected length of life cycle and the expected life cycle maintenance costs. A detailed expression of the expected length of a life cycle is given by Jia and Christer [31]. It is important to mention that the life cycle ends at either a failure or a preventive repair in the case of perfect maintenance. However, the lifetime continues after imperfect maintenance activities and it ends only at a functional failure. Thus, the expected total cost per unit time can be expressed as

$$C(\zeta, x_r) = \frac{TC(\zeta, x_r)}{LC(\zeta, x_r)} \tag{7}$$

Note that the expected total cost of maintenance during a life cycle is

$$TC(\zeta, x_r) = C_f + E[N]C_m + E[N_p]E[C_p] \tag{8}$$

Assuming functional failures are detected only at the moment of condition monitoring, the expected length of life cycle is

$$LC(\zeta, x_r : L) = E[N_p]\zeta \tag{9}$$

where, the cumulative (log) degradation value $L = \{L(t_1), \ldots, L(t_{k-1})\}$ corresponds to the set of monitoring time $T = \{t_1, \ldots, t_{k-1}\}$. Hence the expected number of condition monitoring within a life cycle can be obtained by

$$E[N] = \frac{LC(\zeta, x_r)}{\zeta} \tag{10}$$

To find the expected length of life cycle and the expected number of preventive repairs, two cases that partition the degradation process within a life cycle are discussed. In Case 1, started from time 0, the failure occurs before the first preventive repair (Fig. 1). In Case 2, the system is in a "brand new" condition at time 0. The failure occurs after at least one preventive repair as shown in Fig. 2. The system is periodically monitored and found to be in the normal range. When the degradation is found to be between X_p and X_f, a preventive repair is performed. After this repair, the cumulative degradation is pulled back to some specified degradation level, x_r, and the degradation process in the same life cycle continues. Obviously, there can be any number of such preventive cycles until a failure is detected.

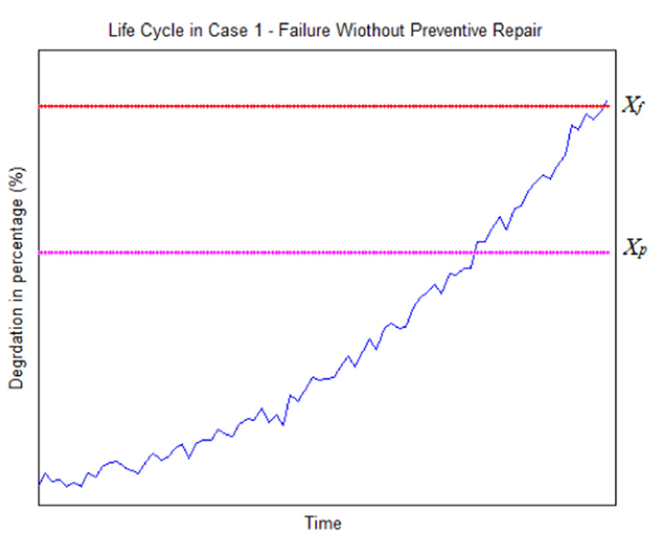

Fig. 1. Case 1; failure occurs before a preventive repair.

3.2. Probabilities of interest

Let D_i denote the monitored cumulative degradation. Given the number of condition monitoring periods within some length of time, i, the length of monitoring interval, ζ, and the initial (log) degradation value, x, the probability that failure occurs before preventive repairs can be given as

$$P_f(i, x, \zeta) = P\{D_1 < x_p, \ldots, D_{i-1} < x_p, x_f < D_i | D_1 = x\} \tag{11}$$

or, equivalently

$$P_f(i, x, \zeta) = P\{D_1 - x < x_p - x, \ldots,\ D_{i-1} - x < x_p - x,\ x_f - x < D_i - x\} \tag{12}$$

Let the increment of the degradation values $\Delta_j = D_j - D_{j-1}$ for $J = 1, \ldots, i$. Then, Eq. (11) can be rewritten as

$$\begin{aligned} P_f(i, x, \zeta) &= P\left\{\Delta_1 < x_p - x, \ldots, \sum_{j=1}^{i-1} \Delta_j < x_p - x,\ x_f - x < \sum_{j=1}^{i} \Delta_j\right\} \\ &= P\left\{x_f - x - \sum_{j=1}^{i-1} \Delta_j < \Delta_i | \sum_{j=1}^{i-1} \Delta_j < x_p - x,\ \sum_{j=1}^{i-2} \Delta_j < x_p - x \ldots, \Delta_1 < x_p - x\right\} \\ &\bullet P\left\{\sum_{j=1}^{i-1} \Delta_j < x_p - x | \sum_{j=1}^{i-2} \Delta_j < x_p - x, \ldots, \Delta_1 < x_p - x\right\} \ldots \\ &\bullet P\{\Delta_1 + \Delta_2 < x_p - x | \Delta_1 < x_p - x\}\ P\{\Delta_1 < x_p - x\} \end{aligned} \tag{13}$$

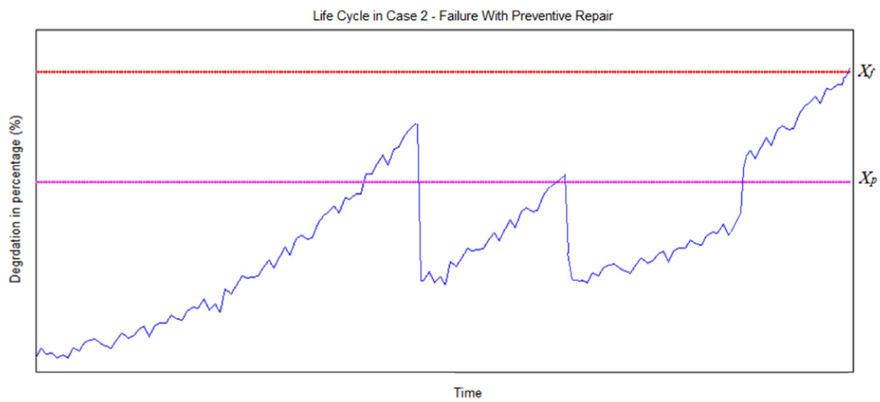

Fig. 2. Case 2; failure occurs after two preventive repairs.

Given $f(\Delta_i)$ as the probability density function (pdf) of the increment, we have

$$P_f(i,x,\zeta)=$$
$$\left(\int_{-\infty}^{x_p-x} f_{\Delta_1}(\zeta_1)\left(\int_{-\infty}^{x_p-x-\Delta_1} f_{\Delta_2}(\zeta_2)\left(\ldots\left(\int_{x_f-x-\sum_{n=1}^{i-1}\Delta_n}^{\infty} f_{\Delta_i}(\zeta_i)d\zeta_i\right)d\zeta_{i-1}\right)\ldots\right)d(\zeta_1\right) \tag{14}$$

Note, $P_f(i,x,\zeta)$ can be obtained with the given distribution of degradation increments. The solution for a uniformly distributed error term is given in Section 5. With partitioning over all possible number of monitoring intervals, the probability that a failure occurs before preventive repairs can be obtained as

$$P_f(x,\zeta)=\sum_{n=0}^{\infty} P_f(n,x,\zeta) \tag{15}$$

The probability that a preventive repair occurs before the failure is

$$P_P(i,x,\zeta)=P\{D_1<x_p,\ldots,\ D_{i-1}<x_p,\ x_p<D_t|D_1=x\}$$

or, equivalently

$$P_P(i,x,\zeta)=P\{D_1-x<x_p-x,\ldots,\ D_{i-1}-x<x_p-x,\ x_p-x<D_i-x<x_f-x\} \tag{16}$$

Similar to the procedure for $P_f(x,\zeta)$, we have

$$P_P(i,x,\zeta)=$$
$$\left(\int_{-\infty}^{x_p-x} f_{\Delta_1}(\zeta_1)\left(\int_{-\infty}^{x_p-x-\Delta_1} f_{\Delta_2}(\zeta_2)\ldots\left(\int_{x_p-x-\sum_{n=1}^{i-1}\Delta_n}^{x_f-x-\sum_{n=1}^{i-1}\Delta_n} f_{\Delta_i}(\zeta_i)d\zeta_i\right)d\zeta_{i-1}\right)\ldots\right)d\zeta_1 \tag{17}$$

Next, $P_P(i,x,\zeta)$ can be obtained using the distribution of degradation increments. By partitioning over all possible number of monitoring intervals, the probability that a failure occurs before preventive repairs can be obtained as

$$P_P(x,\zeta)=\sum_{n=0}^{\infty} P_P(n,x,\zeta) \tag{18}$$

With $P_f(x,\zeta)$ and $P_P(x,\zeta)$ defined above, it is obvious that

$$P_f(x,\zeta)+P_P(x,\zeta)=1 \tag{19}$$

For discrete monitoring mode, in a life cycle, the probability of Case 1 defined earlier is

$P_1=P\{$A failure occurs before the first preventive maintenance$\}$

$$P_1=\sum_{i=0}^{x} P\{D_1<x_p,\ldots,\ D_{i-1}<x_p,\ x_f<D_t|D_1=0\}=P_f(0,\zeta) \tag{20}$$

Similarly, the probability of Case 2 within a life cycle defined earlier is

$P_2=P\{$A preventive repair occurs before a failure$\}$

$$P_2=\sum_{i=0}^{x} P\{D_1<x_p,\ldots,\ D_{i-1}<x_p,\ x_f<D_t|D_1=0\}=P_P(0,\zeta) \tag{21}$$

3.3. *Length of life cycle*

In Case 1, the expected number of preventive repairs and the expected cost of preventive repair are both zero. Hence, the expected length of life cycle for Case 1 is

$$LC(0,\zeta;\text{Case }1)=\sum_{N=1}^{x} N\zeta P_{D_0=0}\{D_1<x_p,\ldots,\ D_{N-1}<x_p,\ x_f<D_N\}$$
$$=\sum_{N=1}^{x} N\zeta P_f(N,0,\zeta) \tag{22}$$

The expected length of life cycle can be obtained given the distribution of degradation increments. In Case 2, the life cycle of system under investigation can be partitioned by a series of preventive cycles (PS) and a failure cycle (FC). Here, a PC is defined as a cycle starting from an initial condition or a just-performed preventive repair and ends when the next preventive repair is performed. A cycle starting from a just-performed preventive repair and ending with a functional failure is called FC. Notice in Case 2, the FC always starts from a performed preventive repair, not the initial condition.

There are two types of PC within a life cycle for Case 2. The first PC starts from degradation value 0 at time 0, and ends with the first preventive repair. Here, this cycle is defined as a beginning preventive cycle (BPC). The PC starting from a performed preventive repair and ending with the next preventive repair is called an intermediate preventive cycle (IPC). For a life cycle in Case 2, the number of IPC could be from 0 to infinity. Obviously, the set of all IPCs within a life cycle in Case 2 starts from the first preventive repair and ends at the last preventive repair before the FC. If only one preventive repair is performed during the lifetime, it is the preventive repair of the BPC and the length of the set of IPCs is 0.

To simplify the notation, in Case 2, denote the IPCs as Cycle 1, and the set of IPCs as Cycle 2, which may contain zero or more IPCs. The FC is denoted as Cycle 3. According to the definitions, in Case 2, the expected length of the life cycle can be presented as

$$LC(x_r,\zeta;\text{Case 2}) = E\left[\text{Cycle 1}\right] + E\left[\text{Cycle 2}\right] + E\left[\text{Cycle 3}\right] \tag{23}$$

Using a similar procedure as in Eq. (21), we have

$$\begin{aligned} E\left[\text{Cycle 1}\right] &= \sum_{k=1}^{\infty} k\zeta P\left\{D_1 < x_p, \dots, D_{k-1} < x_p,\ x_p < D_k < x_f | D_1 = 0\right\} \\ &= \sum_{k=1}^{\infty} (k\zeta) P_P(k,0,\zeta) \end{aligned} \tag{24}$$

$$\begin{aligned} E\left[\text{Cycle 3}\right] &= \sum_{j=1}^{\infty} j\zeta P\left\{D_1 < x_p, \dots, D_{j-1} < x_p,\ x_p < D_j < x_f | D_1 = x_r\right\} \\ &= \sum_{j=1}^{\infty} (j\zeta) P_P(j,x_r,\zeta) \end{aligned} \tag{25}$$

To obtain $E\left[\text{Cycle 2}\right]$, we know that there are $(Np-1)$ IPCs within Cycle 2, and therefor

$$E\left[\text{Cycle 2}\right] = E[\text{IPC}] * (E[Np;\text{Case2}] - 1) \tag{26}$$

Thus, the expected length of IPC is

$$\begin{aligned} E[\text{IPC}] &= \sum_{i=1}^{\infty} i\zeta P\left\{D_1 < x_p, \dots, D_{i-1} < x_p,\ x_p < D_i < x_f | D_1 = x_r\right\} \\ &= \sum_{i=1}^{\infty} (i\zeta) P_P(i,x_r,\zeta) \end{aligned} \tag{27}$$

Given a constant degradation level after each repair, starting from an initial preventive repair, define a success as: a failure occurs before the next preventive repair. Then, the number of preventive repairs after the initial preventive repair follows a geometric distribution, which is

$$(Np-1;\text{Case2}) \sim \text{Geom}\left(P_f(x_r,\zeta)\right) \tag{28}$$

Hence, the expected number of preventive repairs after the initial preventive repair is

$$E[Np;\text{Case2}] = \frac{1}{P_f(x_r,\zeta)} - 1 \tag{29}$$

By Eqs. (26), (27), (29), the expected length of Cycle 2 is

$$E\left[\text{Cycle 2}\right] = \left(\frac{1}{P_f(x_r,\zeta)} - 1\right) \sum_{i=1}^{\infty} (i\zeta) P_P(i,x_r,\zeta) \tag{30}$$

Thus, the expected length of a life cycle for Case 2 can be obtained as

$$LC(x_r,\zeta;\text{Case 2}) = \sum_{k=1}^{\infty} (k\zeta) P_P(k,0,\zeta) + \left(\frac{1}{P_f(x_r,\zeta)} - 1\right) \sum_{i=1}^{\infty} (i\zeta) P_P(i,x_r,\zeta) + \sum_{j=1}^{\infty} (j\zeta) P_P(j,x_r,\zeta) \tag{31}$$

Hence, by Eqs. (22) and (31), the expected length of life cycle is

$$\begin{aligned} LC(x_r,\zeta) &= P_f(0;\zeta) LC(0,\zeta;\text{Case 1}) + P_P(0;\zeta) LC(x_r,\zeta;\text{Case 2}) \\ &= \sum_{N=1}^{x} N\zeta P_f(N,0,\zeta) + \sum_{k=1}^{\infty} (k\zeta) P_P(k,0,\zeta) \\ &\quad + \left(\frac{1}{P_f(x_r,\zeta)} - 1\right) \sum_{i=1}^{\infty} (i\zeta) P_P(i,x_r,\zeta) + \sum_{j=1}^{\infty} (j\zeta) P_P(j,x_r,\zeta) \end{aligned} \tag{32}$$

Now, using Eq. (29), the expected number of preventive repairs within a life cycle can be obtained as

$$E[Np] = P_P(0;\zeta) E[Np;\text{Case 2}] = \frac{P_P(0;\zeta)}{P_f(x_r;\zeta)} \tag{33}$$

Then, the overall expected number of monitoring intervals can be specified as

$$\begin{aligned} E[N] = \frac{LC(x_r,\zeta)}{\zeta} &= (P_f(x_r;\zeta)) \left(\sum_{N=1}^{x} N\zeta P_f(N,0,\zeta) + (P_P(0;\zeta) \right) \\ &\times \left(\sum_{k=1}^{\infty} (k\zeta) P_P(k,0,\zeta) + \left(\frac{1}{P_f(x_r,\zeta)} - 1\right) \sum_{i=1}^{\infty} (i\zeta) P_P(i,x_r,\zeta) \right. \\ &\left. + \sum_{j=1}^{\infty} (j\zeta) P_P(j,x_r,\zeta) \right) \end{aligned} \tag{34}$$

3.4. Life cycle maintenance cost

First, we need to consider the overall expected cost of preventive repairs. With a linear functional relationship, the cost of preventive repairs for Case 1 is 0, and therefore the expected cost of preventive repairs is

$$E[Cp] = M \cdot E[R] + C_s \tag{35}$$

To obtain $E[Cp]$, the expected degradation reduction after a preventive repair, $E[R]$, must be derived. Since the degradation value after a preventive repair is fixed at x_r, the problem reduces to solving the expected degradation level just before a preventive repair. The expected log degradation level just before a preventive repair at time T is

$$E[L(T)] = E\left[\ln\theta + \beta T + \varepsilon(T)\right] = \ln\theta + \beta E[T] + E[\varepsilon(T)] = \ln\theta + \beta E(T) \tag{36}$$

For Cycle 1 in Case 2 we have

$$E[T_1] = E\left[\text{Cycle 1}\right] = \sum_{k=1}^{\infty} (k\zeta) P_P(k,0,\zeta) \tag{37}$$

$$E[L(T)] = \ln\theta + \beta \sum_{k=1}^{\infty} (k\zeta) P_P(k,0,\zeta) - x_r \tag{38}$$

Hence, the expected degradation reduction can be written as

$$E[R_1] = \ln\theta + \beta \sum_{k=1}^{\infty} (k\zeta) P_P(k,0,\zeta) - x_r \tag{39}$$

Similarly, the expected log degradation level just before a preventive repair in the first IPC in Cycle 2 is

$$E[L(T_2)] = L(E[\text{IPC}]) + x_r = \ln\theta + \beta \sum_{k=1}^{\infty} (k\zeta) P_P(k,0,\zeta) + x_r. \tag{40}$$

Thus, the expected degradation reduction is

$$E[R_2] = \ln\theta + \beta \sum_{i=1}^{\infty} (i\zeta) P_P(i,x_r,\zeta) - x_r \tag{41}$$

This expected value is identical for all IPCs in Cycle 2. There is no preventive repair in Cycle 3, according to the definition of FC. Hence, the overall expected value of log degradation reduction just after a preventive repair is

$$E[R] = P_P(0,\zeta)\left(\frac{E[L(T_1)] + (E[N_p]-1)E[L(T_2)]}{E[N_p]}\right)\left(\left(\ln\theta + \beta\sum_{k=1}^{\infty}(k\zeta)P_P(k,0,\zeta) - x_r\right)\right. \tag{42}$$

or, it can be written as

$$E[R] = \left(\frac{P_P(0,\zeta)}{P_f(x_r,\zeta)} - 1\right)\left(\ln\theta + \beta\sum_{i=1}^{\infty}(i\zeta)P_P(i,x_r,\zeta)\right) + P_f(x_r,\zeta)\left(\ln\theta + \beta\sum_{k=1}^{\infty}(k\zeta)P_P(k,0,\zeta) - x_r\right) \tag{43}$$

Then, the expected cost of preventive repairs can be obtained as

$$E[C_p] = \left(\frac{P_P(0,\zeta)}{P_f(x_r,\zeta)} - 1\right)\left(\ln\theta + \beta\sum_{i=1}^{\infty}(i\zeta)P_P(i,x_r,\zeta)\right) + M.P_f(x_r,\zeta)(\ln\theta + \beta\sum_{k=1}^{\infty}(k\zeta)P_P(k,0,\zeta) - x_r) \tag{44}$$

Using Eqs. (6.a), (19), (20), (31)–(33), (42), the expected life cycle maintenance cost is (Eq. (8)) is

$$TC(x_r,\zeta) = C_f + \frac{LC(x_r,\zeta)}{\zeta}C_m + \frac{P_P(0;\zeta)}{P_f(x_r;\zeta)}\left[\left(\frac{P_P(0,\zeta)}{P_f(x_r,\zeta)} - 1\right)\left(\ln\theta + \beta\sum_{i=1}^{\infty}(i\zeta)P_P(i,x_r,\zeta)\right) + M.P_f(x_r,\zeta)(\ln\theta + \beta\sum_{k=1}^{\infty}(k\zeta)P_P(k,0,\zeta) - x_r\Bigg)\right]. \tag{45}$$

Using Eqs. (32) and (45), the expected life cycle maintenance cost per time unit defined in Eq. (7) can be obtained. More details on deriving the cost models can be found in [32,33].

4. Imperfect DBM decision model

In this section, a decision model is proposed to minimize the expected life cycle maintenance cost per unit time. The decision variables, i.e. x_r and ζ, are optimized for the case that the degradation model follows a Markov process. Let denote Z as the value of the life cycle maintenance cost per time unit. Hence, a decision model for imperfect DBM with a linear relationship between the degradation reduction and the cost of a preventive repair can be formulated as

$$\begin{aligned} &\text{Min } Z = C(x_r,\zeta) \\ &\text{s.t.} \quad x_r < X_p \\ &0 < L(t_{i-1}) - x_r \le 1, \quad i = 1,\dots,k. \end{aligned}$$

The constraints imply that the degradation level immediately after a PM cannot be greater than the potential failure threshold. If the expected cost of preventive repairs is a non-linear function of degradation reduction, then a procedure similar to Eqs. (35)–(42) can be followed to derive the decision model (see Section 6).

With the assumption that error terms of the degradation model follow a Markov process, the decision model for periodic monitoring can be optimized through the following steps:

1. Model the degradation or its log transformed form by a random coefficient function with autocorrelated error terms.
2. Estimate the coefficients parameters of the degradation model.
3. With a specified distribution of degradation error increments, solve for the probabilities of interest.
4. Specify the functional relationship and corresponding parameters between expected cost of preventive repairs and degradation reduction.
5. Based on Step 2 and Step 4, specify the objective function of the decision model as a function of monitoring interval and the degradation level after preventive repairs. The constraints of the decision model need to be given with respect to the context of particular applications.
6. Optimize the decision model.

5. Numerical example

For a simplified preliminary study, assume $\mu_\theta = 1$ and $\sigma_\theta^2 = 0$, that is, θ is constant and $\ln\theta = 0$. Set β at 0.005. Assume the error term in the degradation model follows a Markov process and it is uniformly distributed between -0.001 and 0.001. The Markov assumption is particularly reasonable for correlated condition signals caused by short intervals between monitoring such as vibration monitoring. With the above settings, the log degradation function is set with the expected log degradation value proportional to time t.

Based on the above settings, a group of 500 sets of simulated run-to-failure data are generated and then log transformed. The log-transformed form in Eq. (3) is then used to generate degradation data. For vibration data, the thresholds are chosen as follows [34,35]:

- The threshold of failure is set at 0.03 mv, with a corresponding log degradation value, X_f=3.041.
- The threshold of potential failure is set at 0.025 mv, with a corresponding log degradation value, X_p=3.219.

In this numerical example, a linear functional relationship between expected degradation reduction and the cost of preventive repairs is investigated. The cost and coefficient used are configured as Cs=500, and M=2000. If a preventive repair occurs at monitoring i, the linear function in Equation 6a can be specified as

$$C_p[i] = 2000(L(t_{i-1}) - x_r) + 500. \tag{46}$$

The cost values involved are as follows:

- Mean cost per monitoring or inspection; Let C_m=50.
- Mean cost per failure and replacement; Let C_f=50,000.

The monitoring interval is also assumed to be some unknown constants, i.e. $\zeta = (t_i - t_{i-1})$. Since the log degradation error increment is uniformly distributed, it is obvious that

$$\Delta_i = L_i - L_{i-1} = \beta(t_i - t_{i-1}) + (\varepsilon(t_i) - \varepsilon(t_{i-1})) = 0.005\zeta + (\varepsilon(t_i) - \varepsilon(t_{i-1})) \tag{47}$$

where $\Delta_i \sim$Uniform $[0.005\zeta - 0.001,\ 0.005\zeta + 0.001]$, so the probability density functions of Δ_i is

$$f_{\Delta_i}(\gamma_i) = \begin{cases} \frac{1}{0.002}, & 0.005\,\zeta - 0.001 < \gamma < 0.005\,\zeta + 0.001 \\ 0, & \text{else.} \end{cases} \tag{48}$$

Next, $f_{\Delta_i}(\gamma_i)$ needs to be applied to find $P_P(N, x_r, \zeta)$ and $P_f(N, x_r, \zeta)$. Given these probabilities of interest, the expected length of the life cycle is specified by Eq. (32). Accordingly, the expected degradation level just before a preventive repair in Cycle 2 is specified. Consequently, the expected degradation reduction and the expected cost of preventive repairs can be obtained as

$$E[R] = 0.005\left(\frac{P_P(0,\zeta)}{P_f(x_r,\zeta)} - 1\right)\sum_{i=1}^{\infty}(i\zeta)P_P(i,x_r,\zeta))$$
$$+P_f(x_r,\zeta)\left(0.005\sum_{k=1}^{\infty}(k\zeta)P_P(k,0,\zeta) - x_r\right)$$
$$E[C_p] = \left(\frac{P_P(0,\zeta)}{P_f(x_r,\zeta)} - 1\right)\left(\sum_{i=1}^{\infty}(i\zeta)P_P(i,x_r,\zeta)\right)$$
$$+10\,P_f(x_r,\zeta)\left(\sum_{k=1}^{\infty}(k\zeta)P_P(k,0,\zeta) - 200x_r\right) + 500$$

Furthermore, the metrics of interests in the cost model such as the expected number of preventive repairs, the expected life cycle length and the expected number of condition monitoring can be obtained. As a result, the expected total maintenance cost and the expected life cycle maintenance cost per time unit can be calculated. In addition, the decision model to minimize the expected life cycle maintenance cost per time can be formed based on our discussion in Section 4. The values of decision variables at the minimal value of maintenance cost for this example are as follows:

- The optimal monitoring interval for the most cost-effective condition monitoring is $\zeta = 16.143$ time units.
- The optimal log degradation level after each repair is $x_r = 4.111$.

6. Application of methodology on vibration data

This section considers the advancement of the developed models with (i) more general autocorrelation polynomial forms of error terms for the degradation model, (ii) nonlinear functional relationships between the cost of preventive repairs and degradation reduction, and (iii) an optimization approach for degradation models with complex structured autocorrelated error terms. The methodology is applied to a group of simulated vibration signals which are generated with autocorrelated error terms with different variance and covariance structures. The probabilities of interest are obtained based on the frequency count. Also, the expected life cycle maintenance cost is optimized. Finally, a sensitivity analysis is conducted on the cost values and structure parameters of the error terms.

6.1. Degradation data generation and settings

Similar to the numerical example in Section 5, the threshold of failure and the potential failure threshold are set as 0.03 mv and 0.025 mv, respectively [34,35]. The cumulative degradation is expressed as a two-parameter exponential model as in Eq. (3). Similar procedures for data generation and decision model development can be followed for degradation models other than the exponential function. The parameters in the cumulative degradation model for simulation are as follows:

- β=0.005,
- Let $\mu_\theta = 1$ and $\sigma_\theta{}^2 = 0$, that is, θ is constant and $\ln\theta = 0$,
- $\varepsilon(t_i)$ is the error term generated through uni-variate ARMA process with lag=k, k=1,2,…. The error term is centered at 0.

According to the autocorrelation structure, the following settings are discussed:

- *Setting 1: Autocorrelation error at lag=2.*
  Simulated run-to-failure data for 500 groups are generated for the error terms in the degradation model. The cumulative degradation values are obtained by the degradation model with specified parameters for both deterministic and stochastic parts. In this set, the coefficients of the ARMA process for error terms are as follows:
  - AR polynomial coefficient vector: (1, −0.4), and
  - MA coefficient vector: (1, 0.5).

  The procedure for data generation and probability calculation is as follows:
  1. The values of degradation level after preventive repair and the value of the monitoring interval length in a discrete way are x_r=[0, 5, 10, 15, 20] and ζ=[40,80,120,160,200].
  2. H is the number of monitoring intervals from initial degradation 0 to the moment that the first event occurs. We compare the generated degradation values at time $H\zeta$ with the thresholds X_p and X_f.
  3. For each value of H, denoted as h, count the number of degradation values greater than X_p, and count the number of values greater than x_f. The portion with degradation $x > X_f$ out of 500 samples is $P_f(h,\ x_r = 0)$. Fig. 3 shows the histograms of the number of monitoring intervals before the degradation reaches the threshold of failure and potential failure when the initial degradation level is 0. The expected number of monitoring intervals before degradation reaches the threshold of potential failure is tabulated in Table 1a, based on the different values of x_r and ζ. Table 1b contains the expected number of monitoring intervals before degradation reaches the failure threshold.
     From the above tables, it is notable that with the increase of degradation level after each preventive repair, the expected number of monitoring interval is shortened. This is reasonable since the initial degradation value after each preventive repair becomes higher. Moreover, with the increase of the length of monitoring interval, the expected number of monitoring interval is also slightly shortened. This means that a longer length between monitoring would result in some delays in timely preventive activities.
  4. $P_p(\zeta,\ x_r)$ can be obtained by integration over $P_P(k;\zeta,\ x_r)$. In this example, the probability can be obtained directly by summing frequency count. $P_f(\zeta,\ x_r)$ is obtained in a similar way (Table 2). These values are used in the cost model.
     Setting 2: Autocorrelation error at lag=3.
     Simulated run-to-failure data for 500 groups are generated for the error terms in the degradation model. The cumulative degradation values are obtained by the degradation model with specified parameters for both deterministic and stochastic parts. In this set, the coefficients of the ARMA process for error terms are

- AR polynomial coefficient vector: (1, −0.4, 0.8), and
- MA coefficient vector: (1, 0.5, 0.6).

The procedure of data generation and probability calculation is similar to the case with lag=2.

6.2. Optimization of the decision model

The cost and coefficient used are set as Cs=5000, and M=500. The cost values involved are C_m=100, and C_f=50,000. There are two settings in this example as summarized in Table 3.

Given the probabilities of interest for the cost model, the values of the expected number of preventive repairs, the expected life cycle length and the expected number of condition monitoring intervals are calculated based on the procedure developed in Section 3. For each case, the following values need to be determined: the expected life cycle length, the expected number of

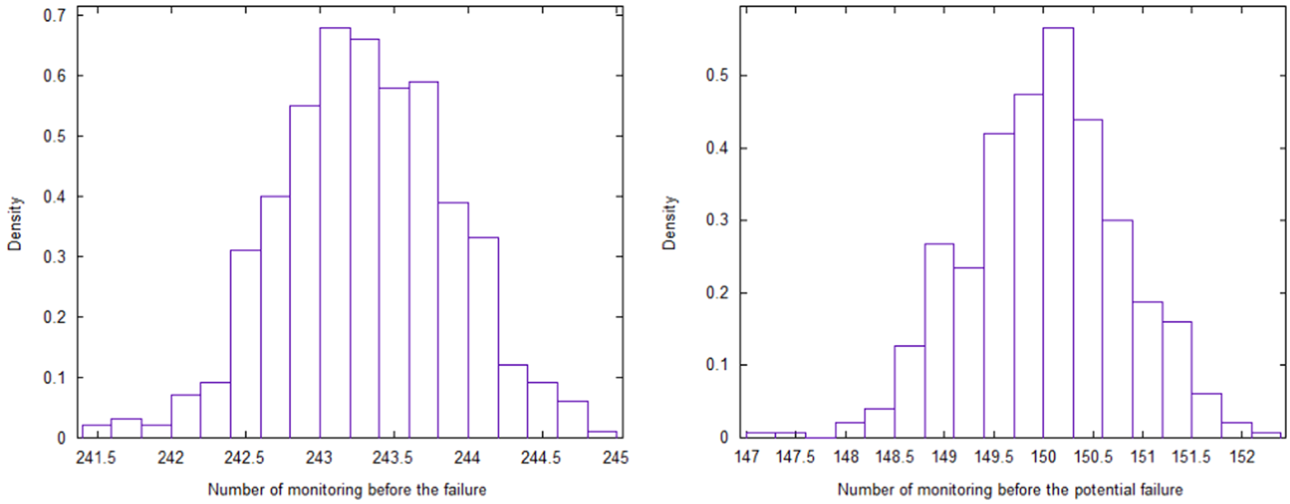

Fig. 3. Number of monitoring intervals.

Table 1
(a) Expected number of monitoring intervals before degradation reaches the potential failure threshold, (b) expected number of monitoring intervals before degradation reaches the failure threshold.

(a)

ζ/x_r	0	5	10	15	50
40	175.003	174.746	167.912	144.540	120.267
80	87.559	87.366	83.912	72.199	59.615
120	58.333	58.253	55.895	48.056	39.619
160	43.750	43.680	41.846	35.959	29.858
200	35.009	34.946	33.506	28.888	23.812

(b)

ζ/x_r	0	5	10	15	50
40	243.287	243.017	236.108	212.812	188.429
80	121.659	121.486	118.017	106.321	93.676
120	81.093	80.999	78.634	70.805	62.379
160	60.827	60.743	58.908	53.013	46.922
200	48.663	48.599	47.145	42.532	37.450

Table 2
Probabilities of interest ($P_f\,(x_r, \zeta)$).

ζ/x_r	0	5	10	15	50
40	0.0270	0.0269	0.0285	0.0320	0.0328
80	0.0268	0.0270	0.0275	0.0312	0.0333
120	0.0270	0.0268	0.0279	0.0322	0.0354
160	0.0269	0.0272	0.0273	0.0295	0.0349
200	0.0268	0.0270	0.0280	0.0316	0.0371

preventive repairs, the expected number of condition monitoring intervals, the expected degradation reduction after a preventive repair, the expected cost per preventive repair, and the expected life cycle maintenance cost per unit time.

The response surface method (RSM) is used to solve the optimization model. The optimal value derived by RSM is not necessarily a global optimal value. However, given appropriate data, RSM is effective in searching heuristic optimal solutions for response with complex forms. Fig. 4 shows the response surface of life cycle per unit time for settings I-A and I-B. Table 4 provides a summary of optimal solutions for all the settings. In this research, vibration signals are abundant and hence the response surface optimization is applicable. In practice, imperfect maintenance strategies based on vibration signals usually lead to discrete degradation reduction solutions at discrete levels including alignment, and replacement of critical parts.

6.3. Sensitivity analysis

Without loss of generality, a sensitivity analysis for setting I-A is given to present the sensitivity of the optimal solution to the changes of coefficients of interest. Table 5 provides a summary of the sensitivity analysis for C_m and C_f.

From the above table, we may observe a trend that with the increase in the condition monitoring cost, the cost effective strategy suggests less frequent monitoring. In other words, the optimal length of monitoring interval increases as the cost of monitoring cost increases. This is reasonable under the objective of minimal cost. On the other hand, as the cost of condition monitoring increases, the degradation level after preventive repairs is lowered. This is because with longer length between monitoring, the risk of failure before taking preventive repairs is increased. Hence, more degradation reduction is needed to reduce this risk. When the criticality of failure events increases, the cost effective strategy recommends us to monitor the system as frequently as possible, i.e. reducing the length of monitoring interval. Again, this is reasonable under the objective of minimal cost. Given a catastrophic cost of failure, a larger degradation reduction is suggested by the optimal solution for minimal cost.

Table 3
Two settings for cost and decision model.

Settings	Error term polynomial specification (lag=2)	Error term polynomial specification (lag=3)
Non-linear Function between preventive repair and degradation reduction (α=3/2) $C_p[i]=500(L(t_{i-1})-x_r)^{\frac{3}{2}}+5000$	Setting 1-A	Setting 2-A
Quadratic Function between preventive repair and degradation reduction (α=2) $C_p[i]=500(L(t_{i-1})-x_r)^2+500$	Setting 1-B	Setting 2-B

Table 4
Summary of optimal solutions.

Setting	I-A	I-B	II-A	II-B
Nonlinear parameter (α)	1.5	2	1.5	2
Order of linear expression error terms (lag)	2	2	3	3
Optimal length of monitoring Interval (hrs)	5.89	6.36	7.905	8.23
Optimal degradation level after maintenance	14.31	13.01	12.41	25

With a higher cost of failure and smaller values for the monitoring interval length, the degradation level after repair must be lower to reduce the cost of unexpected failures.

It is important to emphasize that the sensitivity analysis can be also performed for the parameters in the degradation model. For instance, by increasing the slope of cumulative degradation, the cost effective strategy advocates taking more frequent monitoring. It implies that with rapid increase in degradation, given constant monitoring interval, the risk of failure before taking any preventive repairs is increased. Therefore, more frequent monitoring are needed to reduce this risk. However, the optimal degradation level after each preventive repair is not so sensitive to the change in the slope of cumulative degradation. This is reasonable because the degradation level after preventive repair is affected by elements such as: (i) the uncertainty caused by cumulative degradation patterns, (ii) the cost of preventive repairs, and (iii) the cost of failure.

Table 5
Sensitivity analysis.

C_m	50 (low)	200 (medium)	1000 (expensive)
ζ	6.56	7.93	20
x_r	25	13.11	9.45
C_f	50×10^3 (mild)	200×10^3 (medium)	1000×10^3 (catastrophic)
ζ	7.93	4	4
x_r	13.11	6.01	4.37

7. Concluding remarks and future research directions

A decision model has been developed to minimize the total cost of imperfect DBM by determining an optimal interval of condition monitoring and the degradation level after each preventive repair. The proposed decision model can be applied for any engineering system with traceable degradation measures in which maintenance practices exert considerable influence on the degradation level. The model was successfully applied to a group of simulated vibration signals. The decision model is based on the novel cost model with varying cost of preventive repairs. In this study, both linear and non-linear functional relationships between the expected degradation reduction and the cost of preventive repairs are investigated.

This study has initiated a new area for the research of cost effective maintenance strategies with imperfect preventive repairs. The results clearly indicate the significance of the proposed model and the decision variables under the objective of minimal cost. From the results it is apparent that the optimal length of monitoring interval has a direct relationship with monitoring cost. Moreover, longer monitoring interval increases the risk of failure before taking preventive repairs. Hence, more degradation reduction is needed to reduce this risk. The degradation level after preventive repairs can be lowered by increasing the cost of condition monitoring. Furthermore, with a higher cost of failure, it is required to lower the degradation level after repair to reduce the cost of unexpected failures.

In addition, by increasing the slope of cumulative degradation, the cost effective strategy advocates taking more frequent monitoring. The optimal degradation level after each preventive repair is not so sensitive to the change in the degradation slope mainly due to the uncertainty associated with degradation patterns. Based on this analysis, we may anticipate that the coefficients of error terms in the degradation model would significantly affect the optimal values of degradation level after preventive repairs.

The decision model needs to be extended to incorporate the time for PM or replacement, dependency of the degradations before and after repair, Bayesian estimation of the degradation model parameters, failure to detect the fault, and false alarms. In this research, periodic monitoring mode with constant monitoring interval (ζ) is considered which is quite usual, e.g., in certain power generation systems. However, varying monitoring interval can be studied for future considerations. Further investigation is also required to consider systems with multiple components. This study has developed a maintenance cost model with different functional relationships between the degradation reduction and the cost of preventive repairs. Other functional relationships should be developed to accommodate different practical applications of imperfect maintenance.

Response-surface-based optimization methods, used in this study, require a prior knowledge of the characteristics and shapes of the response model, which creates the need for an abundant data source. In this research, vibration signals are abundant and hence the response surface optimization is applicable. If the observations are not abundant or easy to obtain, other appropriate optimization methods should be applied.

In effect, practical implication of the proposed imperfect maintenance strategy usually leads to discrete solutions. From the perspective of the overall system, the degradation reduction is also available at discrete levels. Hence, given practical concerns, a continuously valued optimal region generated by the proposed decision model that contains discrete valued options might be more appropriate. The choice between several candidate options within a determined optimal region depends on the analysis of characteristics of subsystem to be maintained, the balancing of cost benefits, and industrial requirements.

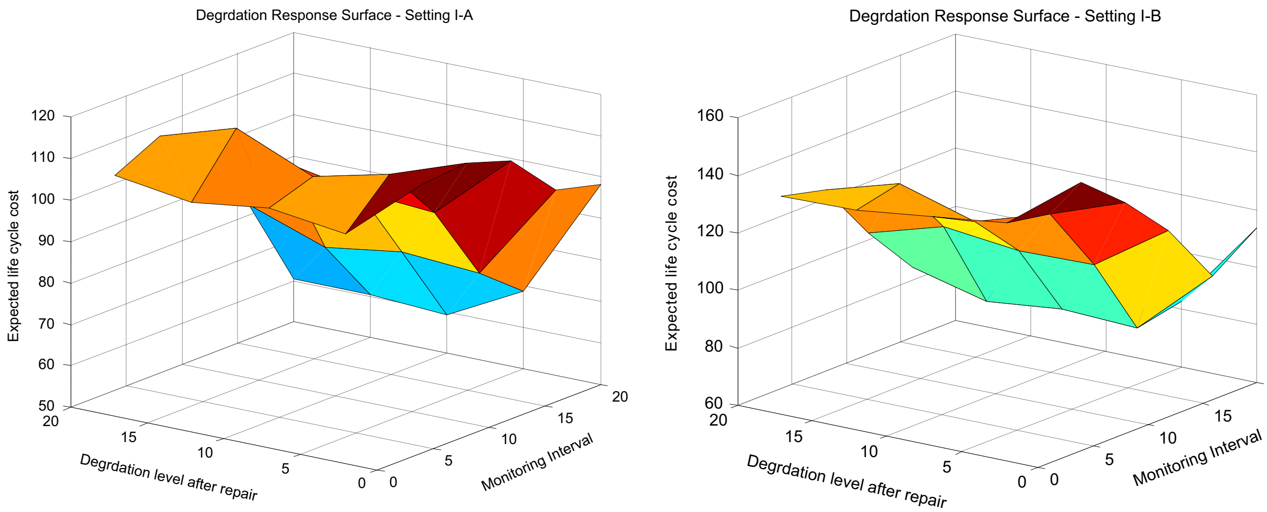

Fig. 4. Response surface of life cycle per unit time for settings I-A and I-B.

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